

Predictive 3C Orchestration for UAV-Swarm Low-Altitude IoT under Environmental Link Uncertainty

First Author, Second Author, and Third Author

Abstract—UAV swarms can support low-altitude IoT services, but their links are affected by mobility, weather, and limited onboard resources. Existing works usually study caching, routing, deployment, or task collaboration separately. This paper presents P3C-Orch, a predictive computation, communication, and caching orchestration framework for heterogeneous UAV swarms under environmental link uncertainty. The framework uses an ANN-based link-risk predictor, cache-value-aware scheduling, and mobility-aware dwell control. The lightweight practical controller, P3C-LR Core, uses predicted link margin and outage risk together with cache value, energy cost, delay cost, load, and handover cost. We also study P3C-SR, an internal stability-regularized variant, and ET-P3C, an exploratory event-triggered ablation. On blind paper-lock seeds across seven regimes, P3C-LR Core ranks first against external baselines. Under combined stress, it improves the aggregate objective by 6.47% over Reactive-3C, reduces average delay by 0.61%, p95 delay by 2.14%, handovers by 24.52%, and outage by 9.14%. These improvements over Reactive-3C are Holm-corrected significant for delay, p95 delay, handovers, and outage. The tradeoff is that P3C-LR Core uses 0.72% more energy and has 1.05% lower useful cache hit ratio than Reactive-3C. P3C-SR achieves the best aggregate non-oracle objective, while ET-P3C does not provide reliable additional gain. The results show that predictive 3C orchestration is useful, but it does not dominate every metric.

Index Terms—Low-altitude economy, UAV swarms, edge intelligence, 3C orchestration, caching, link-risk prediction, mobility-aware scheduling, Internet of Things.

I. INTRODUCTION

Low-altitude economy (LAE) services use UAVs for sensing, delivery, monitoring, and emergency support. In such services, UAV swarms can act as flying edge nodes for IoT users and vehicles. They can relay data, serve local requests, cache popular content, and provide temporary communication coverage. UAV-assisted wireless networks are also a key topic in 5G and beyond systems because UAVs provide flexible deployment and good line-of-sight opportunities [1].

The main difficulty is that the service state changes quickly. Users move, content requests are bursty, UAV energy is limited, and air-to-ground links are affected by distance, altitude, speed mismatch, and weather. A scheduler that only maximizes the current rate may switch swarms too often. A scheduler that only minimizes energy may delay urgent demand. A cache-aware method may still fail when the cached

content cannot be served over a risky link. Hence, computation, communication, and caching, called 3C resources in this paper, should be handled together.

Existing works address important parts of this problem. MUCCO jointly considers coded caching, user grouping, and UAV deployment for energy-efficient data delivery [2]. A recent post-disaster UAV-swarm work uses collaborative beamforming, multi-path routing, and rate maximization for emergency communication [3]. Dynamic task collaboration studies asynchronous two-layer UAV-swarm control and stability for task-volume consensus [4]. These works are useful references, but they do not directly address predictive cache-value-aware 3C scheduling under environmental link uncertainty and competing dynamic content demand.

This paper proposes P3C-Orch, a predictive 3C orchestration framework for heterogeneous UAV swarms. The framework uses an ANN to predict next-slot link margin and converts it into outage risk using weather-wise calibration. It then scores feasible swarm-cluster assignments using predicted link risk, useful cache value, delay, energy, load, fairness, and mobility-aware dwell cost. The main practical method is P3C-LR Core. We also study P3C-SR, an internal stability-regularized variant, and ET-P3C, an exploratory event-triggered ablation.

The contributions are as follows.

- We build a predictive 3C orchestration model for heterogeneous UAV swarms serving moving low-altitude IoT demand under clear, rainy, and rain-hot weather.
- We integrate ANN-based link-risk prediction with weather-wise uncertainty calibration and use it inside the scheduling decision.
- We design P3C-LR Core, a cache-value-aware and mobility-aware controller that jointly considers link margin, outage, energy, delay, load, and handover cost.
- We report P3C-SR as a stability-regularized internal variant and ET-P3C as an exploratory ablation, without hiding negative or mixed results.
- We evaluate all methods on 80 blind paper-lock seeds across seven regimes and report confidence intervals and Holm-corrected statistical tests.

The rest of the paper is organized as follows. Section II reviews related work. Section III gives the system model. Section IV describes the ANN link-risk predictor. Sections V and VI present P3C-Orch and its algorithms. Section VII explains the experimental setup. Section VIII discusses the

The authors are with the Department of Computer Science and Engineering, National Institute of Technology Meghalaya, India.

Implementation availability details are given in the reproducibility note.

results. Section IX gives ablations and limitations. Section X concludes the paper.

II. RELATED WORK

A. UAV Caching and Data Delivery in LAE

Caching can reduce repeated backhaul access and improve latency when content demand is skewed. The coded caching result in [5] provides a basic reference for cache-aided delivery. MUCCO is the closest LAE caching and deployment work for this paper. It jointly optimizes coded caching, user grouping, and UAV deployment, and it uses projected-distance-based grouping, semidefinite programming, and matching theory for energy-efficient data delivery [2]. In contrast, P3C-Orch does not focus on coded caching design. It uses useful cache value as one part of a predictive service-assignment score that also includes link risk, dwell cost, delay, and energy.

B. UAV Swarm Communication and Routing

UAV-swarm communication has been studied for emergency and post-disaster networks. The post-disaster communication work in [3] builds a collaborative self-organizing network using collaborative beamforming, Ford-Fulkerson style routing, and diffusion-model-enabled particle swarm optimization. Its main goal is transmission-rate maximization. This motivates the RateMax baseline in our experiments. Our setting is different because users generate content requests over time and the scheduler must consider cache state, mobility-aware handover, and predicted link risk.

C. UAV Swarm Task Collaboration and Stability

Task collaboration among large UAV swarms is related to orchestration, but it has a different control objective. The work in [4] proposes asynchronous two-layer task collaboration with energy constraints and Lyapunov-based analysis. It focuses on task-volume consensus and hierarchical collaboration. This paper does not claim a new consensus theorem. Instead, P3C-Orch schedules service assignments for communication, caching, and computation. We include P3C-SR as a stability-regularized internal variant, but the final claim is based on measured paper-lock results.

D. Link-Risk Prediction and Environmental Uncertainty

Air-to-ground links depend on altitude, elevation angle, path loss, and line-of-sight probability. We use the standard low-altitude platform path-loss model in [6] as a basis for simulation. UAV energy and movement cost are also important, and UAV communication energy models such as [7] motivate including energy in the scheduler. Our project notes and user presentation reported clear, rainy, and rain-hot link-margin degradation, wider residuals in harsh weather, and link instability due to speed mismatch. Hence, P3C-Orch uses a link-margin predictor and weather-wise risk calibration rather than relying only on current observed margin.

E. Summary of Gap

Table I summarizes the difference. Different from these works, P3C-Orch uses predicted link risk, useful cache value, dwell cost, delay, energy, and outage together for service assignment.

III. SYSTEM MODEL

We consider a time-slotted UAV-swarm service system. Time slots are indexed by $t = 1, 2, \dots, T$. A high-altitude platform or edge controller observes the state of heterogeneous UAV swarms, moving users, content requests, cache states, and weather. It then assigns UAV swarms to active demand clusters.

Let $\mathcal{S} = \{1, \dots, S\}$ denote the set of UAV swarms. Each swarm $s \in \mathcal{S}$ has a position $\mathbf{p}_{s,t}$, bandwidth W_s , transmit power P_s , residual energy $E_{s,t}$, cache $\mathcal{C}_{s,t}$, service capacity, current load $L_{s,t}$, and previous assignments. The altitude is limited to $[h_{\min}, h_{\max}]$. Users and vehicles move according to a Gauss-Markov mobility model, which is commonly used to represent correlated movement in mobile ad hoc simulations [8].

At slot t , active requesting users are grouped into demand clusters $\mathcal{K}_t$. Cluster $k \in \mathcal{K}_t$ has centroid $\mathbf{u}_{k,t}$, demand arrival $A_{k,t}$, deadline $\tau_{k,t}$, priority, requested files, and backlog $Q_{k,t}$. The binary variable $x_{s,k,t} \in \{0, 1\}$ is one if swarm s serves cluster k at slot t . Each cluster is assigned to at most one swarm:

$$\sum_{s \in \mathcal{S}} x_{s,k,t} \leq 1, \quad \forall k \in \mathcal{K}_t. \quad (1)$$

The backlog evolves as

$$Q_{k,t+1} = [Q_{k,t} + A_{k,t} - \mu_{k,t}]^+, \quad (2)$$

where $\mu_{k,t} = \sum_s x_{s,k,t} \mu_{s,k,t}$ is the served traffic and $[y]^+ = \max(y, 0)$. Traffic that remains after the deadline is counted as dropped.

The air-to-ground channel uses distance-based path loss, line-of-sight probability, weather attenuation, speed mismatch penalty, and random shadowing. Let $m_{s,k,t}$ be the observed link margin between swarm s and cluster k . The service estimate is

$$\mu_{s,k,t} = R_{s,k,t} (1 - \hat{p}_{s,k,t}^{\text{out}}), \quad (3)$$

where $R_{s,k,t}$ is the link capacity and $\hat{p}_{s,k,t}^{\text{out}}$ is the predicted outage probability.

The cache value for a swarm-cluster pair is

$$V_{s,k,t}^{\text{cache}} = (1 - \hat{p}_{s,k,t}^{\text{out}}) \left(a_1 H_{s,k,t}^{\text{local}} + a_2 H_{s,k,t}^{\text{nbr}} - a_3 S_{s,k,t} - a_4 N_{s,k,t} \right), \quad (4)$$

where H^{local} is the valid local service hit ratio, H^{nbr} is the neighbor-cache reuse ratio, S is a stale-cache penalty, and N is a neighbor-fetch penalty. This form makes cache useful only when it can reduce delay or energy under the predicted link condition.

The switch cost is

$$C_{s,k,t}^{\text{sw}} = \mathbb{I}\{s \neq s_{k,t}^{\text{prev}}\} (c_0 + c_1 d_{s,k,t} + c_2), \quad (5)$$

TABLE I
COMPARISON WITH CLOSELY RELATED UAV-SWARM AND LAE WORKS.

Work	Main focus	Caching	Routing/communication	Prediction	Mobility/handover	3C orchestration	Main limitation
MUCCO [2]	Energy-efficient data delivery	Yes	Coverage and grouping	No	Mobility-aware deployment	Partial	No ANN link-risk prediction or dwell-aware multi-swarm predictive 3C scheduling
Post-disaster swarm [3]	UAV Transmission-rate maximization	No	Collaborative beamforming and routing	No	Placement optimization	Partial	No cache-value-aware dynamic content scheduling
Dynamic task collaboration [4]	Task collaboration and consensus	No	Control network	No	Two-layer asynchronous control	No	Focuses on task-volume collaboration, not service-level 3C orchestration
P3C-Orch	Predictive service orchestration	Useful cache value	Link-risk-aware assignment	ANN link-risk	Dwell guard	Yes	Simulation-based; real UAV traces still needed

TABLE II
MAIN NOTATION USED IN THE SYSTEM MODEL.

Symbol	Meaning
$\mathcal{S}$	Set of UAV swarms
s	Index of a UAV swarm
$\mathcal{K}_t$	Active demand clusters at slot t
k	Index of a demand cluster
$x_{s,k,t}$	Binary assignment variable
$\mathcal{C}_{s,t}$	Cache state of swarm s
$Q_{k,t}$	Backlog of demand cluster k
$A_{k,t}$	New arrival demand
$\mu_{s,k,t}$	Expected service by swarm s for cluster k
$\hat{m}_{s,k,t+1}$	Predicted next-slot link margin
$\hat{p}_{s,k,t}^{\text{out}}$	Predicted outage probability
$V_{s,k,t}^{\text{cache}}$	Useful cache value
$C_{s,k,t}^{\text{en}}$	Energy cost
$D_{s,k,t}$	Delay cost
$C_{s,k,t}^{\text{sw}}$	Switch/handover cost
$B_{s,k,t}$	P3C-LR Core scheduling score

where $s_{k,t}^{\text{prev}}$ is the previous serving swarm of the persistent cluster and $d_{s,k,t}$ is the distance. No handover is counted for a newly born cluster. Energy and delay are represented as additive costs:

$$C_{s,k,t}^{\text{en}} = C_{s,k,t}^{\text{rx}} + C_{s,k,t}^{\text{move}} + C_{s,k,t}^{\text{fetch}}, \quad (6)$$

$$D_{s,k,t} = D_{s,k,t}^{\text{deadline}} + D_{s,t}^{\text{queue}} + D_{s,k,t}^{\text{miss}} + \hat{p}_{s,k,t}^{\text{out}}. \quad (7)$$

Table II lists the main symbols.

IV. ANN-BASED LINK-RISK PREDICTION

The controller predicts the next-slot link margin before scheduling. For every feasible swarm-cluster pair, the feature vector $\mathbf{z}_{s,k,t}$ includes swarm and cluster identifiers, distance, elevation angle, relative speed, weather state, frequency, bandwidth, transmit power, previous link margin, previous outage flag, cache-hit flag, residual energy, current load, and normalized time. The ANN is an MLP with three hidden layers of 64 ReLU neurons. The output is the predicted next-slot link margin:

$$\hat{m}_{s,k,t+1} = f_{\theta}(\mathbf{z}_{s,k,t}). \quad (8)$$

The margin prediction is converted into outage risk using weather-wise residual calibration:

$$\hat{p}_{s,k,t}^{\text{out}} = \sigma \left(\frac{m_{\text{th}} - \hat{m}_{s,k,t+1}}{\sigma_w} \right), \quad (9)$$

where m_{th} is the outage margin threshold, σ_w is the calibrated residual scale for weather state w , and $\sigma(\cdot)$ is the sigmoid function. This risk value is used in both service estimation and scheduling.

Algorithm 1 ANN-Based Link-Risk Prediction

Require: Feature vector $\mathbf{z}_{s,k,t}$, trained ANN f_{θ} , weather state w , threshold m_{th} , calibration scale σ_w

Ensure: Predicted margin $\hat{m}_{s,k,t+1}$ and outage risk $\hat{p}_{s,k,t}^{\text{out}}$

- 1: Collect distance, elevation angle, relative speed, weather, bandwidth, transmit power, previous margin, previous outage, cache flag, energy, load, and time features.
- 2: Normalize features using the predictor preprocessing rule.
- 3: Compute $\hat{m}_{s,k,t+1} = f_{\theta}(\mathbf{z}_{s,k,t})$.
- 4: Select weather-wise residual scale σ_w .
- 5: Compute $\hat{p}_{s,k,t}^{\text{out}} = \sigma((m_{\text{th}} - \hat{m}_{s,k,t+1})/\sigma_w)$.
- 6: **return** $\hat{m}_{s,k,t+1}$, $\hat{p}_{s,k,t}^{\text{out}}$.

TABLE III
OVERALL ANN LINK-MARGIN PREDICTOR METRICS ON THE PAPER-LOCK PREDICTOR EVALUATION SET.

MSE	RMSE	MAE	R^2	Pearson R	Time/sample (ms)
21.090	4.592	3.865	0.554	0.810	0.0042

TABLE IV
WEATHER-WISE LINK PREDICTOR ERROR AND CALIBRATED RESIDUAL SCALE.

Weather	MSE	RMSE	MAE	R^2	Samples	σ_w (dB)
clear	20.194	4.494	3.910	0.510	1332	4.711
rain_hot	24.200	4.919	3.900	0.512	346	4.310
rainy	21.232	4.608	3.777	0.522	822	4.108

The final paper-lock predictor results are shown in Tables III and IV. The overall RMSE is 4.592 dB and the Pearson correlation is 0.810. Weather-wise errors are not identical. The rain-hot state has the largest RMSE in the paper-lock predictor table, which matches the motivation that harsh weather causes wider residuals.

Fig. 1 shows the predicted and true link margins, and Fig. 2 shows residuals by weather. These figures are used as predictor diagnostics and not as proof that the link model is complete.

V. P3C-ORCH FRAMEWORK

Fig. 3 shows the P3C-Orch system model. The controller receives mobility, cache, load, energy, and link states from the UAV swarms and demand clusters. It predicts next-slot link margin, estimates outage risk, computes cache value, applies dwell control, and assigns each active demand cluster to a feasible swarm.

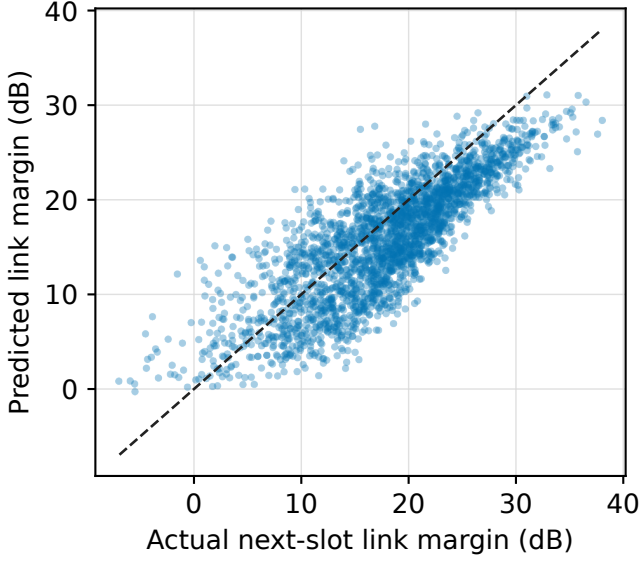


Fig. 1. ANN link-margin prediction scatter on the paper-lock predictor evaluation set. The plot compares predicted and actual margins. The spread shows that prediction is useful but not exact.

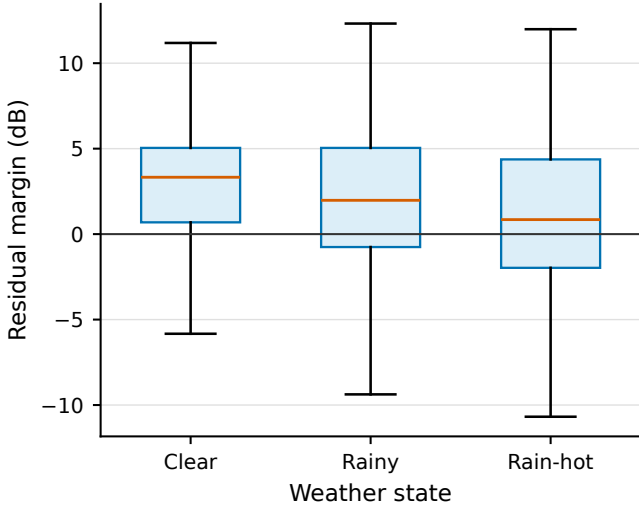


Fig. 2. Prediction residuals by weather state. The rain-hot condition has wider error spread than easier weather states, so outage risk calibration is needed.

For P3C-LR Core, every feasible swarm-cluster pair receives a normalized score:

$$B_{s,k,t} = w_m \tilde{m}_{s,k,t} + w_c \tilde{V}_{s,k,t}^{\text{cache}} + w_f \tilde{F}_{s,t} + w_r \tilde{R}_{s,k,t}^{\text{dwell}} - w_d \tilde{D}_{s,k,t} - w_e \tilde{C}_{s,k,t}^{\text{en}} - w_o \tilde{p}_{s,k,t}^{\text{out}} - w_h \tilde{C}_{s,k,t}^{\text{sw}} - w_l \tilde{L}_{s,t}. \quad (10)$$

Here, $\tilde{m}$ is the normalized risk-adjusted link margin, $\tilde{V}^{\text{cache}}$ is normalized useful cache value, $\tilde{F}$ is a fairness term, $\tilde{R}^{\text{dwell}}$ rewards staying with the previous feasible swarm, $\tilde{D}$ is delay cost, $\tilde{C}^{\text{en}}$ is energy cost, $\tilde{p}^{\text{out}}$ is outage risk, $\tilde{C}^{\text{sw}}$ is switching cost, and $\tilde{L}$ is swarm load. The normalization is done per slot over feasible pairs so that no single raw scale dominates the score.

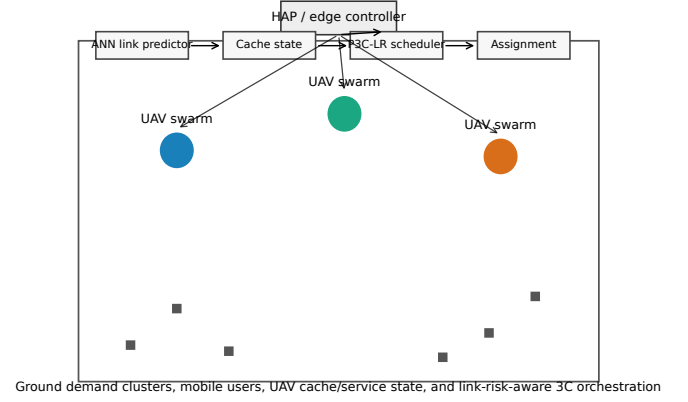
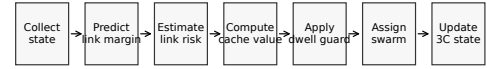


Fig. 3. System model of P3C-Orch. Ground users generate content requests, UAV swarms provide service and cache reuse, and the edge controller uses link prediction, cache state, and scheduling state for assignment.



No Lyapunov stability impulse; P3C-LR ranks feasible assignments by delay, energy, outage, cache value, dwell, and load.

Fig. 4. Algorithm flow of P3C-Orch. The controller collects state, predicts link risk, computes cache value and dwell cost, assigns swarms, and updates queue, cache, energy, and handover states.

The scheduling problem is

$$\begin{aligned} \max_{x_{s,k,t}} \quad & \sum_{k \in \mathcal{K}_t} \sum_{s \in \mathcal{S}} x_{s,k,t} B_{s,k,t} \\ \text{s.t.} \quad & \sum_s x_{s,k,t} \leq 1, \quad \forall k, \\ & \sum_k x_{s,k,t} \mu_{s,k,t} \leq \bar{\mu}_{s,t}, \quad \forall s, \\ & x_{s,k,t} \in \{0, 1\}. \end{aligned} \quad (11)$$

The implementation uses greedy assignment ordered by deadline-aware urgency and backlog for scalability. A dwell guard keeps the previous swarm if it remains feasible and if the new score gain is too small to justify a handover. This guard is important because KMeans-like cluster movement can otherwise create unnecessary handovers.

The paper objective is used only for ranking the final results, not for tuning on paper-lock seeds:

$$\begin{aligned} J = & 0.18 \tilde{D}_{\text{avg}} + 0.16 \tilde{D}_{95} + 0.16 \tilde{E} + 0.14 \tilde{H} \\ & + 0.12 \tilde{p}^{\text{out}} + 0.10 \tilde{P}_{\text{drop}} + 0.08 \tilde{I}_{\text{load}} \\ & - 0.08 \tilde{H}^{\text{cache}} - 0.08 \tilde{T}_{\text{put}} - 0.04 \tilde{F}_{\text{Jain}}. \end{aligned} \quad (12)$$

A lower value of J is better. Individual metrics are always reported with the objective.

Algorithm 2 P3C-LR Core Scheduling

Require: Swarm states, demand clusters, cache states, trained link predictor, dwell parameters

Ensure: Assignment $x_{s,k,t}$ and updated queue, cache, energy, and handover states

- 1: Form active demand clusters and persistent cluster identifiers.
- 2: **for** each feasible pair (s, k) **do**
- 3: Predict link margin and outage risk using Algorithm 1.
- 4: Estimate expected service $\mu_{s,k,t}$.
- 5: Compute useful local and neighbor cache value.
- 6: Compute delay, energy, load, fairness, dwell, and switch costs.
- 7: Normalize all score terms over feasible pairs in the current slot.
- 8: Compute $B_{s,k,t}$ using (10).
- 9: **end for**
- 10: Sort clusters by deadline-aware urgency and backlog.
- 11: **for** each cluster in sorted order **do**
- 12: Select the feasible swarm with the largest score.
- 13: Apply dwell guard; keep the previous swarm when it is feasible and the score gain is small.
- 14: Commit assignment if capacity and energy constraints are satisfied.
- 15: **end for**
- 16: Serve traffic and update $Q_{k,t+1}$ using (2).
- 17: Update cache, energy, load, delay, outage, drop, and handover counters.

VI. P3C-SR AND INTERNAL VARIANTS

P3C-LR Core is the lightweight practical controller. We also study P3C-SR, which adds a stability regularizer to the same predictive 3C score. The regularizer is

$$G_{s,k,t} = U_{k,t}\mu_{s,k,t} - \lambda L_{s,t}\mu_{s,k,t}, \quad (13)$$

where $U_{k,t} = Q_{k,t} / \max(\tau_{k,t} - t + 1, 1)$ is the deadline-aware urgency and $L_{s,t}$ is the swarm load. The P3C-SR score is

$$B_{s,k,t}^{\text{SR}} = B_{s,k,t} + w_q \tilde{G}_{s,k,t}. \quad (14)$$

The term is a regularizer and not the only design objective. It can help when backlog and deadline risk are high, but it can also affect energy, cache use, and throughput. Therefore, we report P3C-SR as an internal variant rather than replacing the practical core.

ET-P3C was also tested as an exploratory event-triggered ablation. It activates a bounded stability impulse only under high-risk states. In the paper-lock results, ET-P3C is close to P3C-LR Core and P3C-SR, but it does not give a clear statistically reliable gain. Hence, ET-P3C is not used as the main method.

VII. EXPERIMENTAL SETUP

The experiment uses the final paper-lock outputs only. The simulator is time-slotted and uses a 1000 m $\times$ 1000 m area with heterogeneous UAV swarms, moving users, dynamic content requests, finite caches, and Markov weather. The altitude range

Algorithm 3 P3C-SR Stability-Regularized Scheduling

Require: Same inputs as Algorithm 2, stability weight w_q , load regularization λ

Ensure: Stability-regularized assignment and updated states

- 1: Build all feasible pair terms as in Algorithm 2.
- 2: **for** each feasible pair (s, k) **do**
- 3: Compute urgency $U_{k,t} = Q_{k,t} / \max(\tau_{k,t} - t + 1, 1)$.
- 4: Compute $G_{s,k,t} = U_{k,t}\mu_{s,k,t} - \lambda L_{s,t}\mu_{s,k,t}$.
- 5: Normalize $G_{s,k,t}$ over feasible pairs.
- 6: Compute $B_{s,k,t}^{\text{SR}} = B_{s,k,t} + w_q \tilde{G}_{s,k,t}$.
- 7: **end for**
- 8: Assign clusters using the same greedy order and dwell guard.
- 9: Update queue, cache, energy, outage, delay, and handover states.

is 100–200 m. Users move with Gauss–Markov mobility and request files from a Zipf-distributed library. Demand clusters are formed from active requesting users.

The evaluated regimes are: R1 normal load, R2 near saturation, R3 overload, R4 severe burst, R5 harsh weather, R6 high mobility, and R7 combined stress. The main paper discussion focuses on harsh weather, high mobility, and combined stress, but all regimes are reported.

The compared methods are Random, Nearest, RateMax, MUCCO-like, Reactive-3C, P3C-LR Core, P3C-SR, ET-P3C, OracleLink Upper Bound, and OracleSelector Upper Bound. OracleLink uses true next-slot link margin. OracleSelector uses hindsight selection. Both are upper bounds and are excluded from practical ranking.

The paper-lock blind seeds are 4001–4080, giving 80 paired runs per regime and method. Earlier seeds from Pilot-0, v2, and v3 are treated only as development evidence. No tuning was run on the paper-lock seeds. The seed audit passed all checks, including seed disjointness, frozen configuration before the run, no tuning, identical seeds/regimes for all methods, and exclusion of Oracle methods from practical ranking. The implementation will be made available at: <https://github.com/Devchandrasen/p3c-orch>.

The primary metrics are average delay, p95 delay, total energy consumption, handovers per 100 active cluster-slots, useful cache hit ratio, outage probability, throughput, and drop rate. Secondary metrics include backlog, SLA violations, load imbalance, fairness, computation overhead, and predictor error. Results are reported as mean $\pm$ 95% confidence interval over the blind seeds when available.

VIII. RESULTS AND DISCUSSION

A. Link-Risk Prediction Performance

The ANN predictor gives an overall RMSE of 4.592 dB, MAE of 3.865 dB, and Pearson correlation of 0.810. Weather-wise RMSE is 4.494 dB for clear, 4.608 dB for rainy, and 4.919 dB for rain-hot. This supports the use of weather-wise calibration, but it also shows that harsh-weather prediction remains imperfect.

B. Main Result Against External Baselines

Table V gives the combined-stress paper-lock results. P3C-LR Core has objective 0.684, while Reactive-3C has 0.731. This is a 6.47% objective improvement over Reactive-3C. P3C-LR Core reduces average delay by 0.61%, p95 delay by 2.14%, handovers by 24.52%, and outage by 9.14%. The corrected p-values are 0.00653 for average delay, 7.60×10^{-7} for p95 delay, 2.53×10^{-7} for handovers, and 0.0285 for outage. These are treated as statistically reliable.

P3C-LR Core does not dominate all metrics. Compared with Reactive-3C, it increases energy consumption by 0.72%, reduces useful cache hit ratio by 1.05%, and increases drop rate by 0.53%. The energy difference is Holm-corrected significant, but it is a worsening for P3C-LR Core. The useful cache and drop-rate differences are not corrected significant in the Reactive-3C comparison.

C. Comparison with MUCCO-like and RateMax

Compared with MUCCO-like, P3C-LR Core improves the objective by 44.32%, average delay by 1.01%, p95 delay by 2.56%, handovers by 83.33%, and useful cache hit ratio by 2.96%. It uses 1.44% more energy, has 11.40% higher outage, and has 2.38% higher drop rate. This shows that MUCCO-like remains strong on energy and drop rate, but it has much higher handover cost in this dynamic setting.

Compared with RateMax, P3C-LR Core improves the objective by 67.20%, average delay by 2.89%, energy by 16.77%, handovers by 91.52%, drop rate by 11.37%, and throughput by 10.27%. However, RateMax has 1.67% better p95 delay and 6.22% lower outage in the combined-stress table. Thus, rate maximization alone is not enough for the aggregate energy-latency-handover objective, but it can still help some tail or outage metrics.

D. Internal Variant Analysis

Table VII compares P3C-LR Core, P3C-SR, and ET-P3C. P3C-SR achieves the best aggregate non-oracle objective, 0.671, and is therefore the strongest overall non-oracle method by the paper objective. P3C-LR Core is the lightweight core and ranks first against external baselines. ET-P3C is close to both methods but does not provide reliable additional gain. The manuscript therefore presents P3C-Orch as a framework containing the practical core and the stability-regularized internal variant.

E. Regime-Wise Behaviour

Fig. 5 shows the delay CDF under the paper-lock evaluation. Figs. 6 and 7 show energy and handover behaviour by regime. P3C-LR Core is useful when link risk and handover decisions matter, but the gains vary by regime. The regime-wise table is generated in `tables/regime_results.tex` and retained for detailed checking.

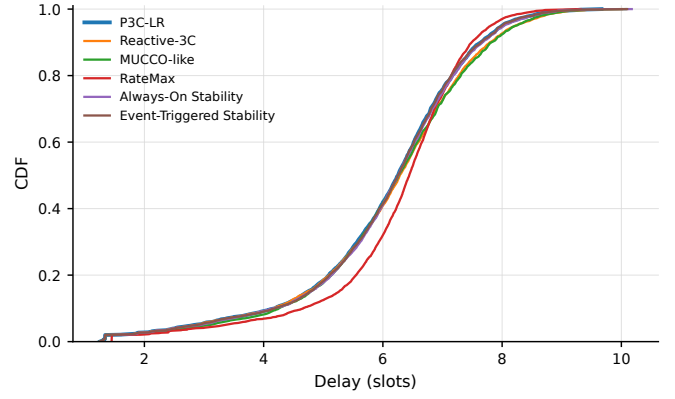


Fig. 5. Delay CDF for the main methods. The figure compares external baselines and internal variants under the final paper-lock evaluation.

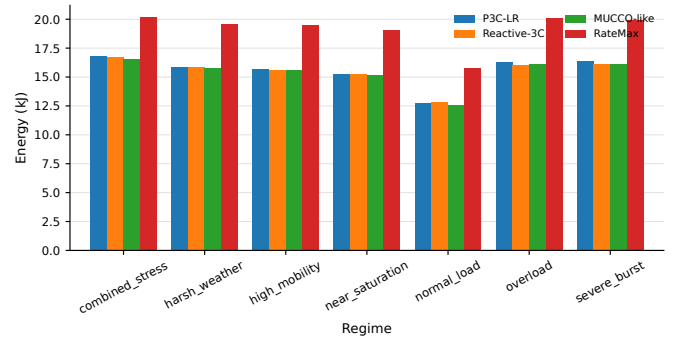


Fig. 6. Energy consumption by regime. P3C-LR Core improves the aggregate objective against external baselines but has a small energy cost compared with Reactive-3C under combined stress.

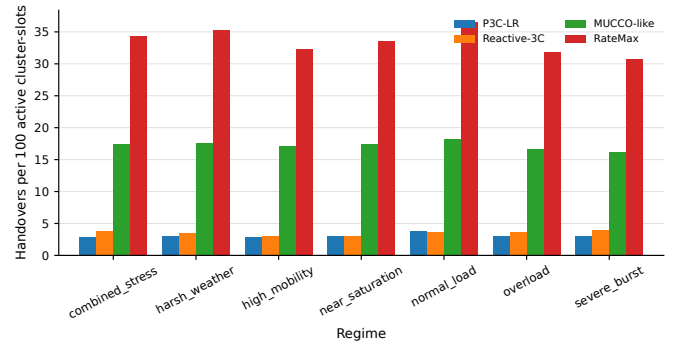


Fig. 7. Handovers per 100 active cluster-slots by regime. Mobility-aware dwell control reduces unnecessary switching compared with rate-centric and caching-centric baselines.

F. Cache, Weather, and Oracle Gap

Fig. 8 shows cache-hit behaviour. Cache value helps when a hit reduces delay or energy, but P3C-LR Core can lose useful cache hit ratio against Reactive-3C because the scheduler also gives weight to outage and dwell. Fig. 9 shows outage by weather. Fig. 10 shows the gap to upper bounds. Oracle methods are not practical baselines and are used only to show remaining headroom.

TABLE V

COMBINED-STRESS PAPER-LOCK RESULTS ON BLIND SEEDS 4001–4080. ENERGY IS IN KJ. CACHE, OUTAGE, AND DROP ARE IN PERCENT. ORACLE METHODS ARE UPPER BOUNDS AND EXCLUDED FROM PRACTICAL RANKING.

Method	Avg Delay	p95 Delay	Energy	Handovers/100	Useful Cache	Outage	Drop	Throughput	Objective
Random	6.32 ± 0.05	7.70 ± 0.08	20.19 ± 0.20	58.15 ± 0.48	0.00 ± 0.00	1.31 ± 0.10	41.55 ± 1.91	598.3 ± 4.7	2.995 ± 0.025
Nearest	6.52 ± 0.04	7.89 ± 0.08	19.50 ± 0.27	1.33 ± 0.16	0.00 ± 0.00	0.23 ± 0.02	42.17 ± 1.82	591.6 ± 8.8	0.732 ± 0.012
RateMax	6.40 ± 0.04	7.72 ± 0.07	20.19 ± 0.20	34.40 ± 0.60	0.00 ± 0.00	0.82 ± 0.06	43.08 ± 1.94	583.5 ± 4.7	2.085 ± 0.024
MUCCO-like	6.28 ± 0.07	8.06 ± 0.09	16.57 ± 0.15	17.49 ± 0.54	26.63 ± 0.62	0.79 ± 0.07	37.29 ± 1.93	653.4 ± 5.7	1.228 ± 0.024
Reactive-3C	6.26 ± 0.07	8.02 ± 0.10	16.69 ± 0.14	3.86 ± 0.22	27.71 ± 0.60	0.96 ± 0.07	37.98 ± 1.91	643.8 ± 5.7	0.731 ± 0.018
P3C-LR Core	6.22 ± 0.07	7.85 ± 0.10	16.81 ± 0.16	2.92 ± 0.22	27.42 ± 0.68	0.88 ± 0.06	38.18 ± 1.84	643.4 ± 6.5	0.684 ± 0.016
P3C-SR	6.24 ± 0.07	7.87 ± 0.10	16.89 ± 0.15	2.77 ± 0.19	27.45 ± 0.63	0.81 ± 0.05	37.82 ± 1.90	647.3 ± 6.2	0.671 ± 0.016
ET-P3C	6.22 ± 0.07	7.84 ± 0.10	16.82 ± 0.15	2.76 ± 0.19	27.44 ± 0.67	0.88 ± 0.06	38.13 ± 1.83	643.4 ± 6.3	0.680 ± 0.017
OracleLink Upper Bound	6.21 ± 0.07	7.85 ± 0.10	16.82 ± 0.15	3.95 ± 0.25	27.58 ± 0.62	0.87 ± 0.06	38.11 ± 1.89	643.9 ± 6.1	0.720 ± 0.019
OracleSelector Upper Bound	6.26 ± 0.06	7.85 ± 0.09	17.96 ± 0.17	11.20 ± 0.74	19.99 ± 0.83	0.73 ± 0.05	39.33 ± 2.00	658.4 ± 4.1	1.020 ± 0.032

TABLE VI

PERCENTAGE CHANGE OF P3C-LR CORE AGAINST EXTERNAL BASELINES UNDER COMBINED STRESS. POSITIVE VALUES FAVOUR P3C-LR CORE; NEGATIVE VALUES SHOW WORSENING.

Baseline	Avg Delay	p95 Delay	Energy	Handovers	Useful Cache	Outage	Drop	Throughput	Objective
Reactive-3C	+0.61%	+2.14%	-0.72%	+24.52%	-1.05%	+9.14%	-0.53%	-0.06%	+6.47%
MUCCO-like	+1.01%	+2.56%	-1.44%	+83.33%	+2.96%	-11.40%	-2.38%	-1.54%	+44.32%
RateMax	+2.89%	-1.67%	+16.77%	+91.52%	n/a	-6.22%	+11.37%	+10.27%	+67.20%

TABLE VII

INTERNAL P3C-ORCH VARIANT COMPARISON UNDER COMBINED STRESS. ENERGY IS IN KJ AND RATIO METRICS ARE IN PERCENT.

Variant	Avg Delay	p95 Delay	Energy	Handovers	Useful Cache	Outage	Drop	Throughput	Objective
P3C-LR Core	6.22	7.85	16.81	2.92	27.42	0.88	38.18	643.4	0.684
P3C-SR	6.24	7.87	16.89	2.77	27.45	0.81	37.82	647.3	0.671
ET-P3C	6.22	7.84	16.82	2.76	27.44	0.88	38.13	643.4	0.680

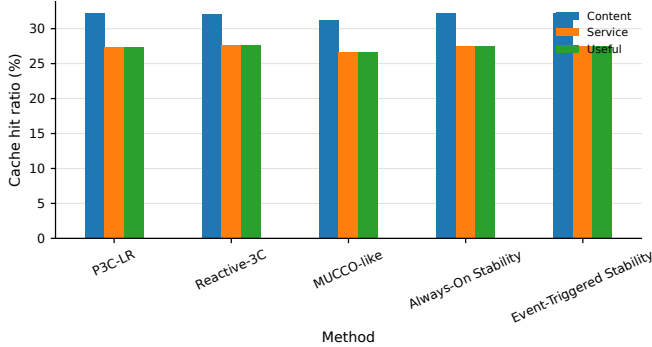


Fig. 8. Cache hit breakdown for the main methods. The figure separates cache effects instead of treating every cached file as useful.

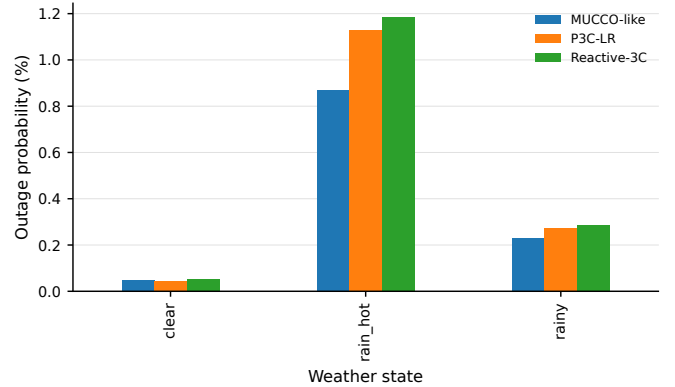


Fig. 9. Outage behaviour by weather. Harsh weather increases link risk, which motivates prediction and calibration.

G. Statistical Significance

Only Holm-corrected significant results are treated as statistically reliable. Table VIII reports the corrected p-values. The main supported improvements over Reactive-3C are average delay, p95 delay, handovers, and outage. Energy is also corrected significant, but it is worse for P3C-LR Core. Hence, the main claim is a latency-handover-outage tradeoff, not universal dominance.

H. Main Observations

- Link-risk prediction is useful when weather and mobility change the service quality.
- Mobility-aware dwell control reduces unnecessary handovers.

- Cache value helps when cached content reduces delay or energy, but cache hit ratio can drop when link-risk and dwell terms dominate.
- P3C-SR gives the best aggregate non-oracle objective, but it does not remove all tradeoffs.
- ET-P3C did not give reliable additional gain in this implementation.

IX. ABLATION STUDY AND LIMITATIONS

A. Ablation Study

Table IX and Fig. 11 show the ablation results. Removing ANN prediction increases delay and p95 delay. Removing cache value removes useful cache hit ratio and increases energy. Removing mobility-aware dwell gives lower average

TABLE VIII

HOLM-CORRECTED WILCOXON TESTS COMPARING P3C-LR CORE WITH EACH BASELINE UNDER COMBINED STRESS. THE YES COLUMN ONLY MARKS CORRECTED $p < 0.05$ AND DOES NOT IMPLY THAT THE DIRECTION FAVOURS P3C-LR CORE.

Baseline	Metric	Corrected p	Effect d_z	$p < 0.05$
Reactive-3C	Avg delay	0.00653	-0.417	Yes
Reactive-3C	p95 delay	7.6e-07	-0.719	Yes
Reactive-3C	Energy	0.0211	+0.404	Yes
Reactive-3C	Handovers	2.53e-07	-0.780	Yes
Reactive-3C	Useful cache	1	-0.161	No
Reactive-3C	Outage	0.0285	-0.382	Yes
Reactive-3C	Drop rate	1	+0.227	No
Reactive-3C	Throughput	1	-0.043	No
MUCCO-like	Avg delay	5.69e-06	-0.701	Yes
MUCCO-like	p95 delay	3.54e-07	-0.761	Yes
MUCCO-like	Energy	5.97e-07	+0.778	Yes
MUCCO-like	Handovers	3.14e-13	-5.492	Yes
MUCCO-like	Useful cache	0.00178	+0.426	Yes
MUCCO-like	Outage	0.0258	+0.304	Yes
MUCCO-like	Drop rate	3.12e-09	+0.918	Yes
MUCCO-like	Throughput	4.03e-09	-0.945	Yes
RateMax	Avg delay	8.04e-09	-0.891	Yes
RateMax	p95 delay	0.0276	+0.412	Yes
RateMax	Energy	3.14e-13	-7.536	Yes
RateMax	Handovers	3.14e-13	-10.877	Yes
RateMax	Useful cache	3.14e-13	+8.806	Yes
RateMax	Outage	0.202	+0.210	No
RateMax	Drop rate	3.14e-13	-3.375	Yes
RateMax	Throughput	3.14e-13	+2.738	Yes
P3C-SR	Avg delay	0.623	-0.246	No
P3C-SR	p95 delay	1	-0.125	No
P3C-SR	Energy	0.0558	-0.368	No
P3C-SR	Handovers	1	+0.145	No
P3C-SR	Useful cache	1	-0.020	No
P3C-SR	Outage	0.024	+0.344	Yes
P3C-SR	Drop rate	0.0313	+0.368	Yes
P3C-SR	Throughput	0.00644	-0.477	Yes
ET-P3C	Avg delay	1	-0.095	No
ET-P3C	p95 delay	1	+0.034	No
ET-P3C	Energy	1	-0.065	No
ET-P3C	Handovers	1	+0.179	No
ET-P3C	Useful cache	1	-0.017	No
ET-P3C	Outage	1	-0.013	No
ET-P3C	Drop rate	1	+0.077	No
ET-P3C	Throughput	1	-0.001	No

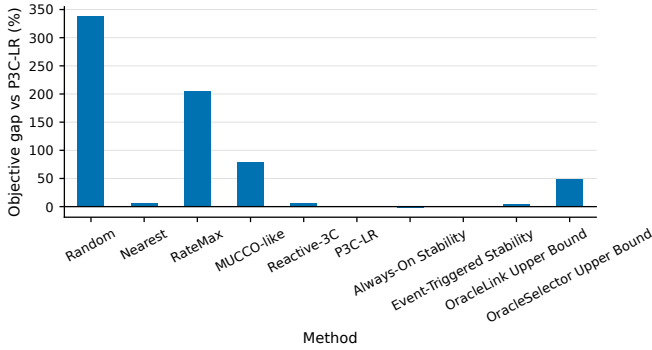


Fig. 10. Oracle gap under the paper-lock objective. OracleLink and OracleSelector are upper bounds and are excluded from practical ranking.

delay in this specific table, but it increases handovers sharply to 24.02 per 100 active cluster-slots. Removing neighbor cache almost halves the useful cache hit ratio. These results support the use of prediction, useful cache value, and dwell control as practical components of P3C-Orch.

B. Stability Branch Analysis

Fig. 12 summarizes the stability branch. P3C-SR is the stability-regularized internal variant and has the best aggregate non-oracle objective. However, earlier stability-focused branches did not give a clean, statistically reliable stability claim across all metrics. ET-P3C is therefore reported as an exploratory ablation. This is why the paper is positioned as predictive 3C orchestration under link uncertainty, not as a pure stability-aware paper.

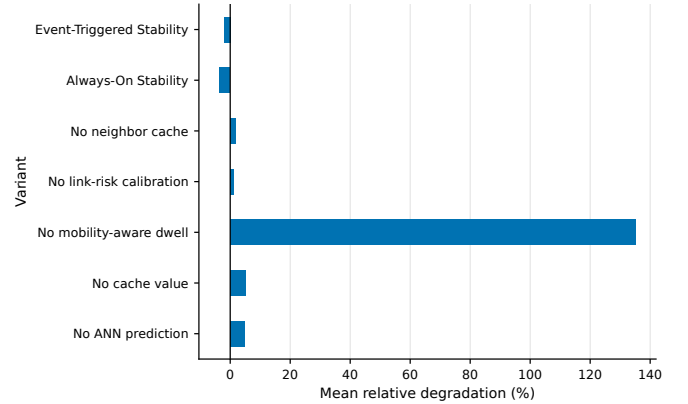


Fig. 11. Ablation results for P3C-Orch. The figure shows the effect of removing prediction, cache value, dwell control, risk calibration, and neighbor cache reuse.

C. Limitations

This work has several limitations. First, the validation is simulation-based. Real UAV tests are needed before deployment. Second, the ANN predictor is trained on generated link data rather than real UAV field traces. Third, the weather model uses simplified clear, rainy, and rain-hot states. Fourth, security and privacy are not considered. Fifth, inter-UAV interference and detailed physical-layer beamforming are simplified. Sixth, P3C-LR Core does not improve every metric; it has a small energy cost and slightly lower useful cache hit ratio and drop-rate performance in some comparisons. Seventh, ET-P3C did not provide reliable gain. Finally, real deployment will need hardware tests, controller delay measurement, and

TABLE IX
ABLATION RESULTS UNDER COMBINED STRESS. ENERGY IS IN KJ AND RATIO METRICS ARE IN PERCENT.

Variant	Avg Delay	p95 Delay	Energy	Handovers	Useful Cache	Outage	Drop	Throughput
P3C-LR Core	6.26	7.94	16.87	2.99	28.16	0.88	38.18	647.3
No ANN prediction	6.33	8.12	16.78	3.65	27.51	0.91	37.87	649.3
No cache value	6.46	8.12	20.21	3.35	0.00	0.79	38.39	645.4
No mobility-aware dwell	6.18	8.02	16.65	24.02	27.69	0.67	38.17	649.3
No link-risk calibration	6.28	8.14	16.64	2.95	28.47	0.96	38.03	644.3
No neighbor cache	6.29	7.94	17.28	3.09	14.21	0.90	38.52	643.3
P3C-SR	6.29	7.97	16.94	2.71	28.21	0.81	37.88	651.6
ET-P3C	6.26	7.91	16.93	2.60	27.80	0.91	38.14	647.1

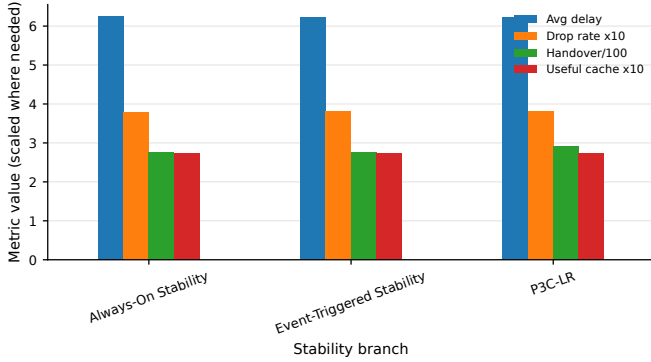


Fig. 12. Stability branch analysis. P3C-SR gives the best aggregate non-oracle objective, while ET-P3C remains an exploratory ablation without a clear reliable gain over the simpler core.

communication-delay validation.

These limitations do not remove the main result, but they define the next steps. Future work should use real link traces, online learning, interference-aware modelling, energy harvesting, and secure orchestration.

X. CONCLUSION

This paper presented P3C-Orch, a predictive 3C orchestration framework for UAV-swarm low-altitude IoT under environmental link uncertainty. The framework combines ANN-based link-risk prediction, cache-value-aware scheduling, mobility-aware dwell control, and optional stability regularization. The practical core, P3C-LR Core, ranks first against external baselines on the paper-lock objective. Against

Reactive-3C, it reduces delay, p95 delay, handovers, and outage, with corrected statistical support for these metrics. It also has tradeoffs: energy consumption is slightly higher and useful cache hit ratio and drop rate are slightly worse. P3C-SR gives the best aggregate non-oracle objective, while ET-P3C is an exploratory ablation without reliable extra gain. Future work will focus on real trace validation, online adaptation, interference-aware link models, energy harvesting, and security-aware service orchestration.

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